# ShelfSense Progress Update

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ShelfSense (CSE499A, Section 13, Group 1) has finished Sprints 1-3, processed 115 new supermarket videos, and trained its Sprint 4 product detector (mAP50 0.913 on SKU-110K). Reading brands on detected product crops doubles the brands the VLM finds compared with whole frames.

## What to show today

Show the evidence trail from raw video to detected products, about 10 minutes in this order. All files are in `E:\ShelfSense`.

1. **The problem (1 min).** Brand companies pay shops for fridge space and placement, but field auditors can be bribed. ShelfSense replaces the auditor's word with video evidence and an automatic audit.
2. **The design (1 min).** Hybrid pipeline: keyframe extraction, then YOLO detects every product, then a small VLM reads brands on crops, then rules compute the KPIs. The VLM alone cannot count (Sprint 3 result).
3. **Keyframe extraction, live (2 min).** Run `python -m src.extraction.extract_frames --video Test_Videos\VID-20261018-WA0092.mp4 --fps 4`, then open its `frames_log.csv` (kept / blurry / duplicate per frame). Show `data\results\figures\blur_check.jpg`: sharp frames kept, smeared ones dropped.
4. **The new footage (1 min).** `data\results\figures\videos_overview_1.jpg` and `_2.jpg`: one frame from each of our supermarket videos.
5. **Baseline VLM failures (2 min).** Sprint 3 table below: the VLM invents brands and answers "10" for every count.
6. **The detector (3 min).** In `data\results\figures`: `sku110k_training_curves.png` (mAP rising over 30 epochs), `prelabel_examples.jpg` (boxes on 12 of our test frames), then `detector_good_and_bad.jpg` to show where it works and where it fails, and crop_brand_examples.jpg for brand reading on crops. Optional live run: `python -m src.detection.prelabel --weights runs\detect\sku110k_v8n\weights\best.pt` (about 1 min).
7. **Tests (30 s).** `python -m pytest -q`: 39 automated tests pass.

## Progress by sprint

Sprints 1-3 are done; Sprint 4′s detector is trained and waiting for corrected labels on our own frames.

| Sprint | Focus | Status | Main result |
|---|---|---|---|
| 1 | Setup and problem definition | Done | Repo, environment, scope, KPIs (cooler purity, paid placement, out-of-stock, share of shelf) |
| 2 | Data and keyframe extraction | Extractor done; tested on 3 fridge clips + 115 supermarket videos | 5,264 frames sampled, 4,558 kept; Crete blur score replaced Laplacian variance |
| 3 | Baseline VLM (Qwen2-VL-2B, zero-shot) | Done on 3 fridge clips | Brand precision 0.13-0.30; every count = 10 or 0; prompts fix format, not perception |
| 4 | Product detection (YOLOv8n) | Detector trained; 222 frames pre-labelled; crop brand test done | See the Sprint 4 section |
| 5 | Integration and rule engine | Not started | |
| 6 | Evaluation and demo app | Not started | |

**Sprint 3 in one line:** two prompt versions cut unparseable answers from 24 to 8, but brands were still invented (Pepsi, 7up in a Coca-Cola fridge) and counts were never real. This is why Sprint 4 adds a detector to do the counting.

## Latest update: 115 supermarket videos

We recorded 115 unique shelf videos (about 22 minutes) in a Bangladeshi supermarket and ran the full extraction on them.

- **Footage:** 119 files, of which 4 were exact copies (found by SHA-256 hash) and 4 were stock clips we excluded. The 111 own clips are 576x1024 portrait, median 11 s.

- **Content:** open shelves, not branded fridges: personal care, biscuits and snacks, noodles, tea and coffee, juices, soft-drink cans, milk, rice and lentils, oil. Price tags (TK) and "OFFER" labels are visible.
- **Split:** 82 train and 29 test videos. Neighbouring clips film the same shelf, so the split keeps blocks of consecutive videos together and the test set never overlaps training.
- **Blur filter, first real test:** the score ranks frames correctly (sharp 0.25-0.35, smeared 0.58-0.62). Near the 0.5 threshold there is no clean gap, so we kept 0.5.
- **Annotation set:** the sharpest 2 frames per video, 164 train and 58 test, chosen automatically.

| Split | Videos | Duration (s) | Frames sampled | Kept | Blurry | Duplicate |
|---|---|---|---|---|---|---|
| Train | 82 | 934 | 3,771 | 3,420 | 54 | 297 |
| Test | 29 | 322 | 1,298 | 1,071 | 115 | 112 |

These videos suit product detection, share of shelf and price tags. They cannot test cooler purity, which still needs branded-fridge recordings.

## Sprint 4: product detector

YOLOv8n trained on 2,000 SKU-110K shelf photos finds products with **mAP50 0.913** (mAP50-95 0.564) on 500 held-out SKU-110K photos.

| Setting (500 SKU-110K val images) | mAP50 | mAP50-95 | Precision | Recall |
|---|---|---|---|---|
| 1024 px, up to 1,000 boxes (chosen) | 0.913 | 0.564 | 0.897 | 0.844 |
| 1024 px, up to 299 boxes | 0.889 | 0.549 | 0.897 | 0.833 |
| 640 px (default size), up to 1,000 boxes | 0.866 | 0.510 | 0.875 | 0.792 |

- **Training:** one class ( `product` , no brands), 30 epochs, 163.5 min on a laptop GTX 1650 (4 GB). mAP was still rising at epoch 30, so more epochs or images should help.
- **Why the settings matter:** one SKU-110K photo holds 718 products and 11 of 500 hold more than 300, so the default 300-box limit costs 2.4 mAP50 points. Small products need 1024 px: 640 px costs 4.7 points.
- **On our 115 videos:** 151,279 products found on 4,558 keyframes, median 29 per frame.
- **Works:** packed shelves at normal distance (toothpaste shelf 63 boxes, drinks shelf 35, nearly every item).

- **Fails:** big products close to the camera. Milk-carton close-ups get a median of 6 boxes per frame, against 53 for canned food. SKU-110K products are small, so the model has not seen close-ups.
- **Not yet measured:** mAP on our own frames. That needs corrected labels; the model has pre-labelled all 222 annotation frames (median 30 boxes each) for correction in Roboflow.

Figures in `data\results\figures`: `sku110k_training_curves.png`, `prelabel_examples.jpg` (12 test frames), `detector_good_and_bad.jpg` (two good shelves, two close-up failures).

## Brand reading: single-product crops vs whole frames

Asking the VLM about one detected product at a time doubles the brands it finds (recall 0.63 vs 0.30) at precision 0.94. This supports the hybrid design chosen after Sprint 3.

**Setup:** Qwen2-VL-2B, fp16, zero-shot (same model as Sprint 3). 12 frames from the 6 drinks-shelf test videos; 120 random detector crops. Answers are matched to a catalogue of the 24 expected brands, as an audit for a brand company would.

| Method (12 frames) | Usable answers | Brand precision | Brand recall |
|---|---|---|---|
| Whole frame, "list the brands" | 6 of 12 (6 looped until the token limit) | 1.00 (on 6 frames) | 0.30 |
| Detector crops, "which brand is this?" | 12 of 12 | 0.94 | 0.63 |

- **Per crop:** 48 of 67 readable crops named correctly (72%; 67% before catalogue matching). Without the catalogue, many answers are near-misses ("chabac", "canad dry", "sprit").
- **Wrong brands are rare but biased:** 2 of 120 crops got a different real brand after matching, both "Coca-Cola" (for Canada Dry and Sprite). This is the Sprint 3 pull towards famous brands, so violations still need a human check of the evidence crop.
- **No answer:** 11 readable crops, mostly Thai or Bengali script (Doi Kham, Latina) and vertical text.
- **Logos:** on crops with no legible text the VLM still named Gatorade (from its G logo) and Fanta correctly.

Caveats: crop and frame labels were drafted by Claude and need a team check; 6 videos show patterns, not significance; the catalogue was built from the test labels, so it is an upper bound. On this 4 GB laptop each crop takes 21 s (model partly in CPU memory).

Figure: `data\results\figures\crop_brand_examples.jpg`.

## Next steps

1. Correct the detector's pre-labels on our 222 frames in Roboflow and draw empty gaps. Two team members check each test frame, since these are the ground truth.
2. Fine-tune the detector on our 164 train frames and report mAP and counting error on our 58 test frames (script ready: `src/detection/finetune_ours.py`).
3. Verify the crop and frame brand labels (team check), then build the per-client brand catalogue into the pipeline and test crops from fridge videos for cooler purity.
4. Record branded beverage fridges (15-30 s each, original quality), including some with competitor products, to test cooler purity.
5. Sprint 5: connect extraction, detection and VLM into one pipeline that writes the JSON audit report.

**Open issues:** the Sprint 3 ground truth is still a draft (fridge_03 brands unverified). The video categories were labelled from one frame each and need a team check. SKU-110K's license allows academic use only, so its photos are never shared.